How mobile networks work

What really happens when a call is made from one mobile network to another? Instead of boring infographics, we try and explain it through the game of football instead...

1 Imagine a call being initiated, just like a goalie looking to pass the ball to his player

2 When the number is dialled, the phone sends the request to the cell tower it's currently connected to – much like the goalie passing the ball to his defender to take forwards.

3 The call request is forwarded to the base transceiver station (BTS) from the tower. This is like Xavi making a pinpoint pass through the midfield to another Barcelona player.

4 From the BTS, the call is connected to the base station controller (BSC), which allots the call a frequency channel to communicate over. This is akin to a football team keeping their shape, and making sure they stay in their positions (or their allotted "frequency").

5 The call is now switched to a mobile switching centre (MSC). If the call is for another user in the same home network, the steps are retraced to the base station of the other user. This is like reaching the goal mouth of the opponents' goal and back passing all the way back to your own goal line.